9 Compare Linear and Polynomial Regression using Python
Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.pipeline import make_pipeline
from sklearn.metrics import mean_squared_error

X = np.array([[1], [2], [3], [4], [5]])
y = np.array([1, 4, 9, 16, 25])

linear = LinearRegression()
linear.fit(X, y)

poly = make_pipeline(
    PolynomialFeatures(degree=2),
    LinearRegression()
)
poly.fit(X, y)

print("Linear Prediction:", linear.predict([[6]])[0])
print("Polynomial Prediction:", poly.predict([[6]])[0])

print("Linear MSE:", mean_squared_error(y, linear.predict(X)))
print("Polynomial MSE:", mean_squared_error(y, poly.predict(X)))
```

Output:

